BUSINESS ANALYTICS / MONTE CARLO SIMULATION

Uncertainty, Pricing, and Store-Level Profitability: A Three-Scenario Monte Carlo Analysis of Starbucks

Spreadsheet-based decision analysis

Research Portfolio Report

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Abstract

This study uses a hypothetical **Starbucks** store to investigate **how uncertainty in customer demand, inflation, and mobile-app adoption interacts with cost and pricing assumptions to shape monthly profitability**. Three deliberately separate Monte Carlo models evaluate (1) a **supply-chain cost shock** with exogenously reduced demand, (2) **inflation** transmitted to prices and procurement costs with an elasticity-based demand response, and (3) a threshold staffing response to **app adoption** followed by cost-proportional pricing. Each scenario contains 1,000 simulated draws. The workbook's saved results imply mean modeled monthly profits of **$8,356.03, $4,225.71, $34,753.67**, respectively. The first two scenarios use a **$15,000 baseline** including procurement cost; the third uses a **$37,000 labor-only baseline.** These figures are **scenario-specific** and are not estimates of actual Starbucks earnings. The analysis illustrates how apparently favorable **operational changes** can have countervailing effects, and it separates stochastic outcome variability from Monte Carlo estimation error.

Keywords: Monte Carlo simulation; price elasticity; stochastic demand; inflation; digital adoption; decision analysis

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1. Introduction and Business Context

Coffeehouse operations face simultaneous uncertainty in input costs, consumer traffic, selling prices, and service processes. **Supply-chain disruptions** may raise the cost of purchased goods; **inflation** can alter both unit costs and prices; and **digital ordering** can change how labor is deployed. These mechanisms matter for profitability because revenue and cost need not move in the same direction. A **decision model** can help isolate each channel before a manager attempts to combine them into a more realistic integrated forecast.

Starbucks is a useful business context because its FY2022 Form 10-K describes Mobile Order & Pay, Starbucks Rewards, digital adoption, and investments in store productivity. Its September 2022 Reinvention announcement also discusses operational efficiency, store formats, and digital convenience [1, 2]. Those documents establish the relevance of the questions; they do not provide the store-level parameters used in this study. The hypothetical **$4 price, 10,000-cup** monthly baseline, distributions, staffing threshold, and elasticity are analytical assumptions, not observed company estimates.

The analysis asks three questions: **How vulnerable is monthly profit to a procurement shock when demand is independently uncertain? How does inflation change profit when price-sensitive customers adjust their purchases?** And **can mobile-app adoption improve profit through labor savings when those savings are fully passed through to prices?** The three models answer different conditional questions; they are not sequential forecasts of one store.

2. Modeling Framework

2.1 Baselines and comparability

Simulations 1 and 2 begin with P0 = $4 per cup, Q0 = 10,000 purchased cups, and SC0 = $2.50 per cup. Their baseline revenue is $40,000; procurement cost is $25,000; and baseline modeled profit is OR = $15,000. **Simulation 3** begins with the same price and demand but includes only $3,000 monthly labor cost. Its baseline modeled profit is $37,000. No corporate overhead, tax, capital expenditure, or other operating costs are represented. Hence these are simplified contribution-style model outputs, not GAAP operating profits.

Scenario	Uncertainty mechanism	Modeled cost	Profit baseline
1 · Supply shock	Normal demand; fixed supply shock	Purchased inventory cost	$15,000
2 · Inflation	Uniform inflation; elastic demand	Purchased inventory cost	$15,000
3 · App adoption	Beta adoption; clipped Normal labor cost	Labor cost only	$37,000

2.2 Simulation 1: supply-chain disruption

The model imposes a **20% increase in procurement cost** and a 20% increase in the selling price: SC = $3.00 and P = $4.80. Demand is independently drawn as D = INT(MAX(0, Normal(8000,1000²))). Sales equal MIN(D,10,000); revenue equals P × sales; procurement cost equals 10,000 × SC = $30,000; and profit equals revenue minus procurement cost. Unsold units do not generate revenue, but their entire acquisition cost is charged in the modeled month. This is a deliberately simplified expense-recognition convention, not

standard inventory accounting. The lower demand distribution is imposed separately; it is not inferred as the causal effect of the price increase.

2.3 Simulation 2: inflation and demand elasticity

A random **inflation rate r ~ Uniform(0.02,0.05)** multiplies both price and procurement cost after the same 20% supply shock. Thus P = 4 × 1.2 × (1+r) and total procurement cost = 10,000 × 2.5 × 1.2 × (1+r). Demand uses the signed linear elasticity approximation D = INT[10,000 × {1 + (−1.2) × (P−4)/4}]. Sales are MIN(D,10,000), revenue is P × sales, and profit is revenue minus the full procurement cost. Conditional on inflation, demand is determined by price; there is no additional independent Normal demand shock in this scenario. The **elasticity formula** is a local percentage-change approximation rather than a constant-elasticity demand curve.

2.4 Simulation 3: mobile adoption, staffing, and pass-through

App adoption A follows Beta(5,3), whose theoretical mean is 62.5%. A staffing reduction is feasible only when **A > 0.50**. Otherwise labor cost L stays at $3,000. Above the threshold, draw X ~ Normal(2000,500²) and set L = MIN(3000,MAX(0,X)). This clipping operation differs from resampling a truncated normal: observations beyond the boundaries are assigned to the nearest boundary.

Define the labor-cost reduction fraction rL = (3000−L)/3000. The assumed selling price is P = 4 × (1−rL) = 4L/3000. Demand follows D = MAX(0,INT[10,000 × {1 + (−1.2) × (P−4)/4}]). The 10,000-cup figure is a baseline, not a capacity limit: all demand is assumed fulfilled. Revenue = P × D, modeled cost = L, profit = revenue − L, and profit change is measured relative to $37,000. App adoption is not a direct multiplier of sales or revenue; it affects outcomes through the staffing threshold. This strong full cost-to-price pass-through assumption is a scenario mechanism, not a description of actual Starbucks pricing policy.

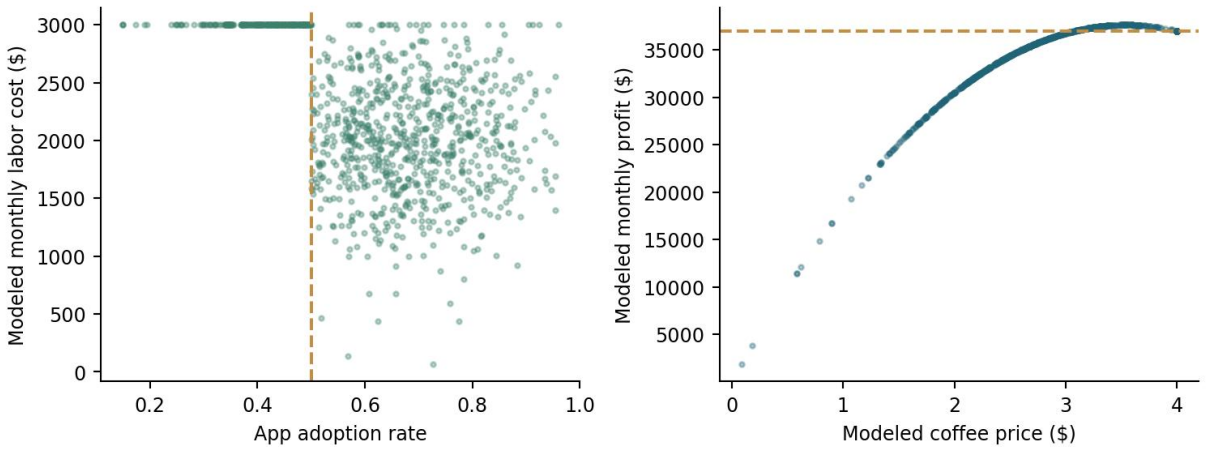


Figure 1. App adoption threshold, modeled labor cost, price, and profit.

Points reproduce the saved 1,000-draw Simulation 3 worksheet values.

3. Simulation Design and Reproducibility

Each core model contains **1,000** worksheet-level Monte Carlo observations generated with volatile RAND-based formulas and distribution transforms. Scenario outputs are calculated row by row, then aggregated into mean values, empirical percentile intervals, and comparisons with scenario-specific baselines. Simulation 3 additionally contains 12 independent 1,000-draw batches on a dedicated replication worksheet, using

ordinary spreadsheet formulas. A replication-row mean summarizes 1,000 draws; it must not be interpreted as a month of observed operating performance. Separate illustrative 12-month scenarios are independent single-draw monthly outcomes without trend, seasonality, or inventory carryover.

For a sample mean, the approximate **95% Monte Carlo confidence interval** is mean ± 1.96 × s/√n, where s is the sample standard deviation across the n simulated observations. A 95% outcome interval instead spans the empirical 2.5th and 97.5th percentiles of individual draws. The former measures numerical uncertainty in the estimated mean; the latter describes dispersion of possible modeled monthly outcomes. Neither interval incorporates uncertainty about whether the assumed distributions or behavioral mechanisms are correct.

All reported numerical summaries and figures use the cached 1,000-row values saved in the final portfolio workbook, not independently generated replacement samples. Recalculation in Excel changes RAND draws, so reported values represent a reproducible *saved-output snapshot*, not immutable constants. To reproduce these exact numbers, retain a frozen copy of the displayed simulation results; to study Monte Carlo stability, recalculate repeatedly or compare independent batches.

4. Results

Scenario	Mean profit	Vs. baseline	P(profit < baseline)	95% outcome interval
1 · Supply shock	$8,356.03	-44.3%	91.3%	$-410.52 to $17,228.16
2 · Inflation	$4,225.71	-71.8%	100.0%	$3,213.54 to $5,149.25
3 · App adoption	$34,753.67	-6.1%	55.9%	$24,844.70 to $37,621.38

Table 2. Descriptive results from 1,000 saved draws per scenario. Different baselines and modeled costs prohibit interpreting cross-scenario profit levels as like-for-like operating performance.

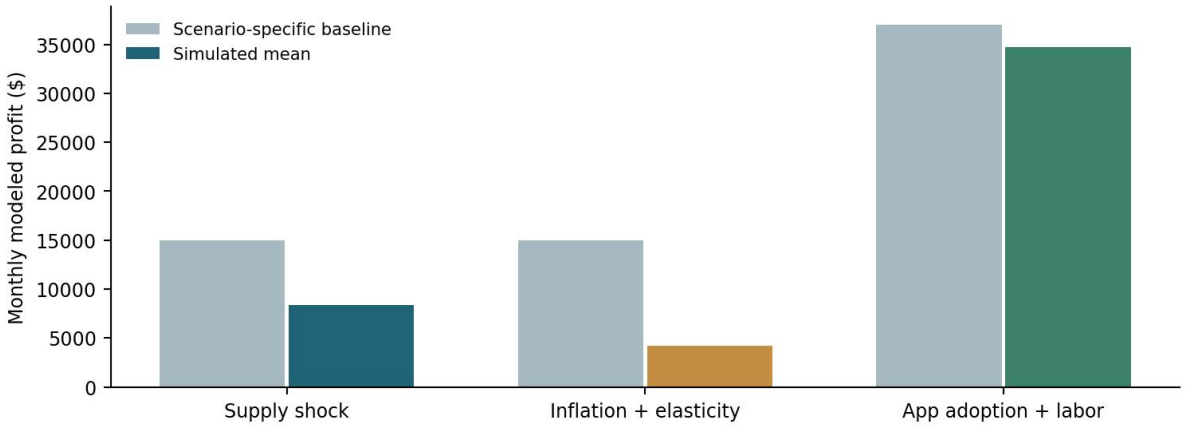


Figure 2. Scenario-specific modeled baseline and simulated mean profit.

The labor-only baseline in Simulation 3 is not comparable to the procurement-cost baselines.

4.1 Supply-chain shock

In the saved simulation, **mean monthly profit** is **$8,356.03**, **-44.3%** relative to the $15,000 baseline. The estimated probability of profit below baseline is 91.3%. The outcome interval ($-410.52 to $17,228.16) is much wider than the Monte Carlo mean interval, approximately $8,069.45 to $8,642.62. Fixed procurement

of 10,000 units makes cost insensitive to unsold inventory; consequently low demand translates directly into weaker profit despite a higher unit price.

4.2 Inflation with elastic demand

The saved **mean profit** is **$4,225.71**, or **-71.8%** versus its $15,000 baseline. The simulated profit-below-baseline frequency is 100.0%, with a 95% outcome interval of $3,213.54 to $5,149.25. The price increase raises nominal unit revenue, but the signed elasticity approximation reduces the quantity purchased while procurement cost increases. Because inflation is the sole independent random driver in this model, the outcome distribution is relatively concentrated compared with Simulation 1.

4.3 App adoption and labor-cost savings

The saved **mean app adoption** is **62.3%**, and 77.2% of draws exceed the 50% staffing threshold. Average modeled labor cost is $2,233.02, average price $2.98, average demand 13,067.5 cups, and **mean monthly profit $34,753.67**. This is **-6.1%** relative to the $37,000 labor-only baseline. The frequency of profit below baseline is 55.9%. **The lower labor bill does not automatically translate into higher profit** because the model passes the cost saving proportionally to customers through lower prices; the associated demand response may not offset lost revenue per cup.

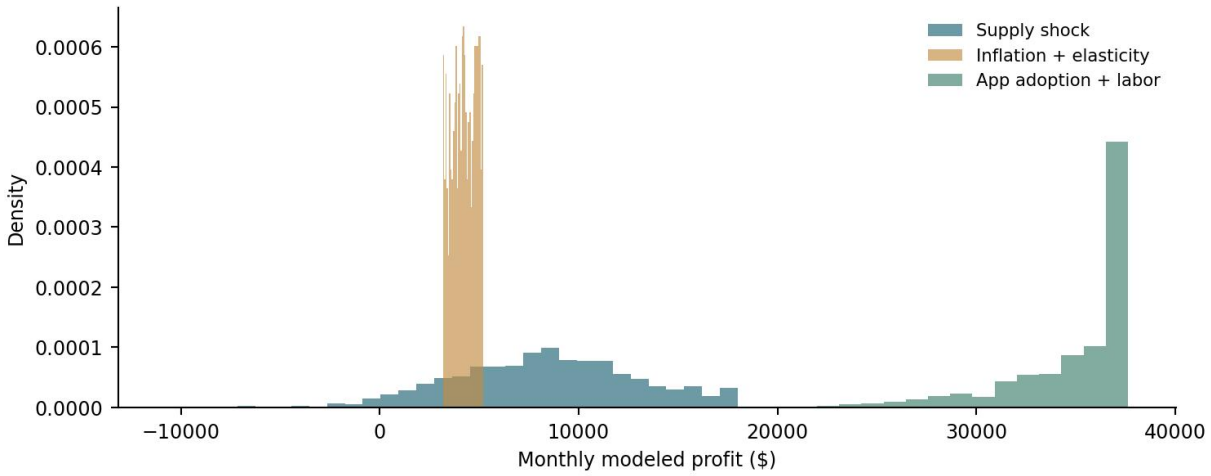


Figure 3. Empirical profit distributions from the saved simulation draws. Density curves reflect different scenario assumptions and should not be compared as identical cost definitions.

5. Convergence and Numerical Precision

Running means offer a practical convergence diagnostic: early sample averages move considerably, while later observations generally have less influence on the cumulative estimate. Under independent draws with finite variance, the standard error of a simulated mean declines approximately with $1/\sqrt{n}$. Thus increasing sample size from 100 to 1,000 reduces standard error by a factor of about $\sqrt{10}$, all else equal. A stable trace supports Monte Carlo numerical precision, but it does not establish that the demand, elasticity, adoption, or cost assumptions match the real world.

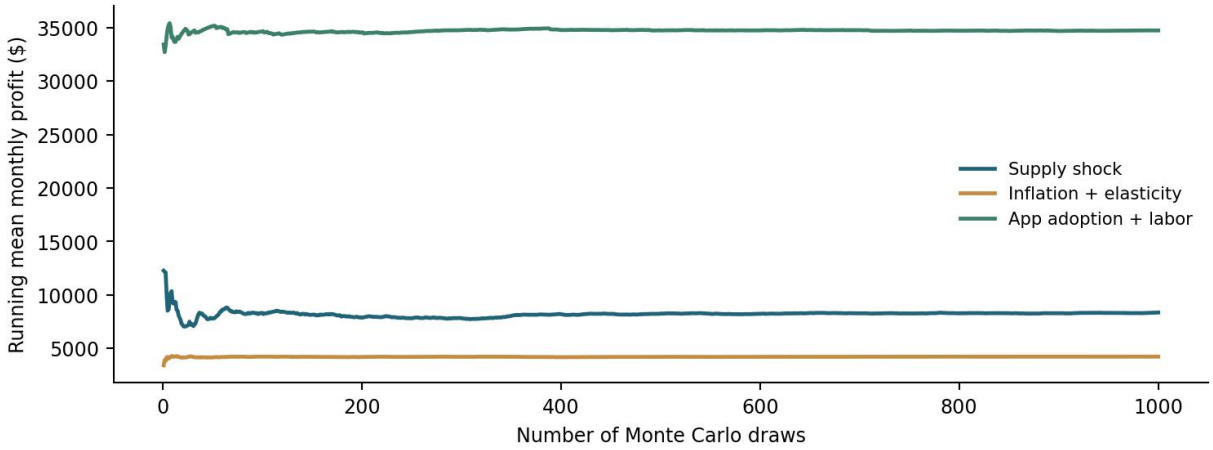


Figure 4. Cumulative mean profit over the 1,000 saved observations in each scenario.

Different vertical levels reflect different cost bases.

Approximate 95% mean-estimation intervals from the saved sample are: Simulation 1, $8,069.45–$8,642.62; Simulation 2, $4,189.53–$4,261.90; and Simulation 3, $34,514.13–$34,993.20. These intervals condition on the model and treat the simulated draws as independent. The Simulation 3 batch replications provide an additional practical check on how much the estimated mean changes across independently recalculated groups.

6. Managerial and Analytical Implications

First, managers should distinguish a unit-price increase from an improvement in contribution. Simulation 1 demonstrates that a higher selling price can coexist with lower profit when purchased inventory remains fixed and demand weakens. Before increasing price or ordering inventory, a business would ideally estimate demand elasticity, waste, stockouts, and salvage value rather than assume a fixed purchase commitment.

Second, the inflation scenario illustrates the importance of **jointly modeling demand and costs**. Price pass-through can protect nominal revenue per item while reducing sales volume. A real decision analysis would vary the assumed elasticity and compare alternative pricing policies, rather than treating −1.2 as a universally valid constant.

Third, the app scenario separates **digital adoption** from labor savings. A high adoption rate only permits a lower-cost staffing regime under the assumed threshold; **it does not directly guarantee a productivity gain**. Moreover, a firm need not pass 100% of a labor saving into price. Management would want to test partial pass-through, capacity constraints, service quality, employee scheduling, app maintenance cost, and the potential benefits of convenience or customer retention. These are recommendations for subsequent analysis, not additional results from the current workbook.

7. Limitations and Research Extensions

This study is a scenario-based teaching and portfolio exercise, not an empirical causal evaluation of Starbucks operations. The parameters are illustrative and not fitted to transaction-level or payroll data. Simulation 1 imposes lower demand independently of price and expenses all purchased inventory immediately. Simulation 2 uses a linear elasticity approximation and a single random inflation factor. Simulation 3 uses a discontinuous 50% threshold, a clipped labor-cost distribution, complete cost-to-price

pass-through, and unlimited demand fulfillment. In particular, zero clipped labor cost implies a zero coffee price under the pricing rule, a mathematical boundary rather than a credible business policy.

Additional work could estimate store-level demand from observed transactions, calibrate adoption from app usage records, test sensitivity to price elasticity and staffing thresholds, and incorporate inventory, service capacity, labor regulations, operating expenses, and implementation costs. Such changes would require a new model specification and should not be retroactively attributed to the current simulation.

8. Conclusion

The **three scenarios** demonstrate a central principle of business simulation: **the effect of an operational shock depends on the complete chain linking uncertain inputs, managerial rules, customer responses, and cost accounting**. In the saved workbook snapshot, all three models yield mean profits below their respective simplified baselines, but the magnitudes are not directly comparable because the third model uses labor cost alone. The strongest insight is methodological: clearly separating assumptions, stochastic mechanisms, Monte Carlo estimation uncertainty, and managerial interpretation prevents a numerical simulation from being mistaken for an empirically validated forecast.

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